TWISTED CONJUGACY IN IRREDUCIBLE SPHERICAL ARTIN GROUPS

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Abstract. We prove that the Artin groups of types D_5, E_6, E_7, E_8, F_4, H_3 and H_4 have property R_∞. Together with previously known cases, this shows that every irreducible Artin group of spherical type and rank ≥ 2 has R_∞.

The proof uses the characteristic pure subgroup and its abelianization, viewed as the permutation lattice on the reflections. Rational, absolutely irreducible constituents of multiplicity one yield a fixed vector after composing with a suitable inner automorphism. The finite action that must be controlled is governed by automorphisms of Coxeter groups, so no description of $\mathrm{Aut}(A)$ is required. The constituents are explicit: eigenspaces of strongly regular graphs for E_n, frame modules for F_4 and H_3, and matrix coefficients in the icosian model for H_4.

The same argument gives uniform proofs in types A_n, B_n and D_n; for D_5 this case appears to be new.

[Contribution statement — placeholder, to be finalised by the authors.] This work was developed with substantial assistance from AI systems (Anthropic's Claude) for the mathematical argument, the programs, the Lean formalisation and the text. So far it has been reviewed only by AI systems: an internal audit, two rounds of an AI "cold referee", and an external AI review of the manuscript. None of this is journal peer review, and no human expert has yet checked the work. The authors must replace this paragraph with an accurate statement of contributions before submission or posting.

1. Introduction

Let G be a group and $\varphi \in \mathrm{Aut}(G)$. Elements $x, y \in G$ are *φ-twisted conjugate* if $y = zx\varphi(z)^{-1}$ for some $z \in G$. The number $R(\varphi) \in \{1, 2, \ldots, \infty\}$ of classes is the *Reidemeister number*, and G has *property* R_∞ if $R(\varphi) = \infty$ for every $\varphi \in \mathrm{Aut}(G)$ [17, 15].

For an Artin group $A = A(\Gamma)$ with Coxeter group $W = W(\Gamma)$, the question "which Artin groups have R_∞?" is Problem 8.6 of the AIM problem list on Artin groups [1]. Known cases include:

- braid groups [14] and pure braid groups [9];
- the spherical types A_n, B_n, D_4, $I_2(m)$ [5];
- D_n for $n \geq 6$ [30, Thm. 1];
- many non-spherical types [24, 30, 29].

The case D_5 was left open in [30]. The authors write there: "The only remaining case in the family $A[D_n]$ is $A[D_5]$. At present we do not have a description of $\mathrm{Aut}(A[D_5])$ ...; we conjecture though that both $A[D_5]$ and $A[D_5]/Z(A[D_5])$ have property R_∞" [30, §1]. Paris–Soroko later computed $\mathrm{Aut}(A[D_5])$ [28], but they do not discuss R_∞. (The survey list in [29, §1] attributes D_n, $n \geq 5$, to [30]; as far as we can see, D_5 is not proved there.) For spherical types the known proofs use an explicit description of $\mathrm{Aut}(A)$ or $\mathrm{Aut}(A/Z(A))$. For $I_2(m)$ they use instead the virtual freeness of $A/Z(A)$ [5].

Main Theorem. *The Artin groups of types D_5, E_6, E_7, E_8, F_4, H_3 and H_4 have property R_∞.*

For $E_6, \ldots, H_4$ see Section 4. For D_5 see Proposition 5.2 and Remark 5.3, where we also note a short alternative derivation from [28, Thm. 1].

Corollary. *Every irreducible Artin group of spherical type and rank ≥ 2 has property R_∞.*

Date: Draft, November 2026.

The corollary follows from the Main Theorem together with [5] (types $A_n, B_n, D_4, I_2(m)$) and [30, Thm. 1] (D_n, $n \geq 6$). Rank one must be excluded: $A(A_1) \cong \mathbb{Z}$, and inversion on $\mathbb{Z}$ has Reidemeister number 2. The method also gives uniform proofs for A_n, B_n and D_n (Section 5), so among the irreducible spherical types only $I_2(m)$ needs a different argument (Remark 5.5).

Significance and novelty. For the six exceptional types $\mathrm{Aut}(A)$ is not known (cf. [28, §1]). The general criteria based on hyperbolic actions of $\mathrm{Aut}(A)$, or on Aut-invariant quasimorphisms [18, 19, 31], also require control of $\mathrm{Aut}(A)$.

The Main Theorem would follow from Conjecture (NA) of [30, §6]: finitely generated, residually finite, non-amenable groups have R_∞. This appears as a theorem in [16], but its proof is reported to be flawed [30, §6]. Our results verify (NA), and Conjecture (R) [25, Problem 19.28], for these groups.

The broad strategy of obtaining an eigenvalue one via representation theory has precedents:

- Dekimpe–Gonçalves–Ocampo [9], for pure braid groups;
- Dekimpe–De Rock–Penninckx [8, Thm. 4.7], for flat manifolds. They require a $\mathbb{Q}$-irreducible $\mathbb{Z}$-subrepresentation η_0 of the holonomy representation whose image is not $\mathbb{Q}$-conjugate to that of any other $\mathbb{Q}$-subrepresentation (in particular, it has multiplicity one), and such that for every D_0 in the integral normaliser of $\eta_0(F)$ some $\eta_0(A)D_0$, $A \in F$, has eigenvalue one;
- the determinant criterion of Dekimpe–Penninckx for crystallographic groups [11, Cor. 3.10], as stated in [10, Thm. 3.3];
- the eigenvalue-one property of finite groups announced by Hiss–Lutowski [23]. For odd-dimensional constituents this gives an alternative route to the witnesses.

What is new here is:

- the application to Artin groups through the *characteristic reflection lattice* $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$;
- the control of the relevant finite symmetries by $\mathrm{Aut}(W)$ [21];
- the explicit sign and swap witnesses;
- the completion of the exceptional cases, and of D_5.

Overview. Let $P = \ker(A \to W)$, let $\mathcal{T}$ be the set of reflections, and let $\rho\colon W \to \mathrm{Sym}(\mathcal{T})$ be the conjugation action.

- P is characteristic, and $H_1(P;\mathbb{Z}) \cong \mathbb{Z}^{\mathcal{T}}$ is a permutation module (Section 2).
- An automorphism φ acts on $\mathbb{Z}^{\mathcal{T}}$ as πN, where π realises the induced automorphism of $\rho(W) \cong W/Z(W)$ and N commutes with $\rho(W)$. On a good constituent U, $N = \pm 1$.
- For a suitable inner automorphism ι_a, the composite $\varphi' = \iota_a \circ \varphi$ has a fixed vector. So $R(\varphi'|_P) = \infty$, hence $R(\varphi') = \infty$, and $R(\varphi) = R(\varphi')$ (Section 3).
- The possible π are controlled by $\mathrm{Aut}(W)$. Explicit constituents and witnesses finish the proof (Section 4).

Formal verification. A Lean 4/Mathlib formalisation accompanies this paper. It contains no `sorry`, no `axiom` and no `native_decide`. The Lean kernel checks:

- the Reidemeister lemmas and the criterion (both cases);
- the Artin group as a presented group, the map f, and the equivariant surjection $P^{ab} \to \mathbb{Z}^{\mathcal{T}}$;
- for each exceptional type, finite certificates (commutant-scalar constituents and ± 1 witnesses).

The cited inputs are explicit hypotheses: W is finite; P is characteristic; $P^{ab} \cong \mathbb{Z}^N$ as groups, with $N = |\mathcal{T}|$ (Brieskorn); and a statement `hOut` that every automorphism of $W/Z(W)$ is inner, or inner times a given π_0. `hOut` is proved in Lean for H_3 and F_4, and is a hypothesis for H_4, E_6, E_7, E_8 (cf. Appendix B). The formalisation predates the use of [21] below and the explicit constituents of Section 4; it uses computer-found constituents. Type D_5 is not formalised.

2. The characteristic reflection lattice

Lemma 2.1. *Let $\psi \in \mathrm{Aut}(G)$.*

(a) If $K \trianglelefteq G$ with $\psi(K) = K$, then $R(\psi) \geq R(\bar{\psi})$.
(b) If $G = \mathbb{Z}^m$ and ψ is given by B, then $R(\psi) = |\operatorname{coker}(I - B)|$. This is ∞ if B has eigenvalue 1.
(c) $R(\iota_g \circ \psi) = R(\psi)$, where $\iota_g(x) = gxg^{-1}$.
(d) If $[G : H] < \infty$ and $\psi(H) = H$, then $R(\psi|_H) \leq [G : H]\, R(\psi)$.

Proof. (a) The projection maps classes onto classes.
(b) $y \sim x$ iff $y - x \in (I - B)\mathbb{Z}^m$.
(c) $x \mapsto xg$ maps $\iota_g\psi$-classes bijectively to ψ-classes.
(d) Write $G = \bigsqcup_i Hg_i$. If $y = zx\psi(z)^{-1} \in H$ with $z = hg_i$, then y is $\psi|_H$-conjugate to $g_i x \psi(g_i)^{-1}$. □

Notation. For a Coxeter graph with vertex set S:

- $A = \langle S \mid \Pi(s, t; m_{st}) = \Pi(t, s; m_{st})\rangle$, where $\Pi(s, t; m) = sts\cdots$ has m letters;
- $W = A/\langle\langle s^2 \rangle\rangle$, with quotient map $\mu\colon a \mapsto \bar{a}$, and $P = \ker \mu$;
- $\mathcal{T}$ is the set of reflections, $\rho(w)(r) = wrw^{-1}$, and $\rho(w)$ also denotes the permutation matrix on $\mathbb{Z}^{\mathcal{T}}$.

Since $\mathcal{T}$ generates W, $\ker \rho = Z(W)$. Each $\alpha \in \mathrm{Aut}(W)$ induces θ_α on $\rho(W)$ by $\theta_\alpha\rho(w) = \rho(\alpha w)$. For finite irreducible W of rank ≥ 3, $Z(W) \subseteq \{\pm 1\}$, and $r \mapsto \rho(r)$ is injective on $\mathcal{T}$.

Theorem 2.2 ([7, Thm. 1.4])**.** *If A is irreducible of spherical type, then P is characteristic in A.*

This was extended to reducible types in [20]. The proof in [7] is computer-assisted in several cases:

- for H_4, F_4, E_6, E_7, E_8 it uses GAP calculations [7, Prop. 5.4, Cor. 5.5];
- for H_3 it uses one GAP calculation [7, Lemma 5.1];
- among the classical types it uses GAP for B_4, B_6, D_4, D_6 [7, Lemmas 3.5, 4.4].

We have not re-run these calculations. Each $\varphi \in \mathrm{Aut}(A)$ thus induces $\varphi_W \in \mathrm{Aut}(W)$ and $\theta_\varphi := \theta_{\varphi_W}$.

Lemma 2.3. *For any Coxeter group, $s \mapsto (e_s, s)$ defines a homomorphism $f\colon A \to \mathbb{Z}^{\mathcal{T}} \rtimes W$, with $(v, w)(v', w') = (v + w \cdot v', ww')$ and $w \cdot e_r = e_{wrw^{-1}}$.*

Proof. For $\Pi(s, t; m)$, the $\mathbb{Z}^{\mathcal{T}}$-component is $\sum_{k=1}^{m} e_{r_k}$ with $r_k = (st)^{k-1}s$. For $\Pi(t, s; m)$ it is $\sum_k e_{r'_k}$ with $r'_k = (st)^{m-k}s$. Both sets equal $\{(st)^j s : 0 \leq j < m\}$. Compare [7, Lemma 3.7]. □

Proposition 2.4. *Let A be of spherical type. Then $f(P) = 2\mathbb{Z}^{\mathcal{T}}$, and f induces an isomorphism $H_1(P; \mathbb{Z}) \cong 2\mathbb{Z}^{\mathcal{T}}$ under which conjugation by a acts as the permutation $\rho(\bar{a})$.*

Proof. We have $f(as^2a^{-1}) = (2e_{\bar{a}s\bar{a}^{-1}}, 1)$, and these elements generate $f(P)$. Moreover $P \cong \pi_1$ of the complexified reflection-arrangement complement [2], whose H_1 is free on the meridians, one per hyperplane [3, 27]. So $H_1(P; \mathbb{Z})$ has rank $|\mathcal{T}|$, and the surjection onto $2\mathbb{Z}^{\mathcal{T}}$ is an isomorphism. □

Corollary 2.5. *Let M_φ be the matrix of φ_* on $\mathbb{Z}^{\mathcal{T}}$. Then $M_\varphi \rho(w) M_\varphi^{-1} = \theta_\varphi(\rho(w))$ and $M_{\iota_a \circ \varphi} = \rho(\bar{a})M_\varphi$. Hence θ_φ preserves the permutation character $\chi_{\mathcal{T}}(g) = \#\{r : g(r) = r\}$.*

The image $f(A) \cong A/[P, P]$ is the crystallographic quotient of [22, 26]; see also [13].

3. The fixed-vector criterion

A $\rho(W)$-invariant subspace $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is *good* if $U \otimes \mathbb{C}$ is irreducible and occurs with multiplicity one in $\mathbb{C}^{\mathcal{T}}$. A good U is a full isotypic component.

Lemma 3.1. *If U is good and $N \in GL(\mathbb{Z}^{\mathcal{T}})$ commutes with $\rho(W)$, then $NU = U$ and $N|_U = \pm 1$.*

Proof. $NU \cong U$, so $NU = U$. By Schur's lemma $N|_U = c \in \mathbb{Q}^\times$. Applying $N^{\pm 1}$ to a primitive vector of $U \cap \mathbb{Z}^{\mathcal{T}}$ gives $c^{\pm 1} \in \mathbb{Z}$. □

Lemma 3.2. *A nonzero invariant $U \subseteq \mathbb{Q}^{\mathcal{T}}$ is good in each of the following cases.*

(i) Every orbital matrix of $\rho(W)$ on $\mathcal{T} \times \mathcal{T}$ acts on U by a scalar.
(ii) $\langle \chi_U, \chi_U \rangle = \langle \chi_U, \chi_{\mathcal{T}} \rangle = 1$.
(iii) $U \otimes \mathbb{C}$ is irreducible and $\sum_r \dim(U \otimes \mathbb{C})^{\mathrm{Stab}(r)} = 1$, where r runs over representatives of the $\rho(W)$-orbits on $\mathcal{T}$ (Frobenius reciprocity).

Proof. (i) The orbital matrices span the commutant $\mathcal{C} = \mathrm{End}_{\mathbb{C}W}(\mathbb{C}^{\mathcal{T}}) = \prod_i M_{m_i}(\mathbb{C})$, where $\mathbb{C}^{\mathcal{T}} = \bigoplus_i V_i \otimes \mathbb{C}^{m_i}$. By hypothesis $\mathcal{C}$ maps $U_{\mathbb{C}} = U \otimes \mathbb{C}$ into itself and acts on it by scalars. The central idempotents $e_i \in \mathcal{C}$ act on $U_{\mathbb{C}}$ by 0 or 1 and sum to 1, so exactly one e_i acts as 1. Hence $U_{\mathbb{C}} \subseteq V_i \otimes \mathbb{C}^{m_i}$ meets only one isotypic component. Being W-invariant, $U_{\mathbb{C}} = V_i \otimes L$ for some $0 \neq L \subseteq \mathbb{C}^{m_i}$, and $M_{m_i}(\mathbb{C})$ preserves L and acts on it by scalars. This forces $m_i = 1$ and $L = \mathbb{C}$. (ii) and (iii) are standard. □

Theorem 3.3. *Let A be irreducible of spherical type. Suppose that for every $\varphi \in \mathrm{Aut}(A)$ there is a permutation π of $\mathcal{T}$ with $\pi\rho(w)\pi^{-1} = \theta_\varphi(\rho(w))$ for all w, such that one of the following holds.*

(i) (stable) *There is a good U with $\pi U = U$ such that, for each $\varepsilon = \pm 1$, some $h\pi$ ($h \in \rho(W)$) has eigenvalue ε on U.*
(ii) (swap) *There are good $U_1 \neq U_2$ with $\pi U_1 = U_2$ such that, for each $\varepsilon = \pm 1$, some $(h\pi)^2$ has an eigenvector in U_1 with eigenvalue ε.*

Then A has property R_∞.

Proof. $N := \pi^{-1} M_\varphi$ commutes with $\rho(W)$.

Case (i). $N|_U = c = \pm 1$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U$ with $h\pi v = cv$, and let $\varphi' = \iota_a \circ \varphi$. Then $M_{\varphi'} v = h\pi N v = v$.

Case (ii). $N|_{U_i} = c_i$. Choose $h = \rho(\bar{a})$ and $0 \neq v \in U_1$ with $(h\pi)^2 v = c_1 c_2 v$. With $Q = h\pi$ we have $QU_1 = U_2$, and $M_{\varphi'}^2 v = QNQNv = v$. Since $M_{\varphi'} v \in U_2$ and $U_1 \cap U_2 = 0$, $u = v + M_{\varphi'} v$ is a nonzero fixed vector.

So $M_{\varphi'}$ fixes a nonzero vector of $\mathbb{Z}^{\mathcal{T}}$. By Lemma 2.1(b),(a) and Proposition 2.4, $R(\varphi'|_P) = \infty$. Then $R(\varphi') = \infty$ by (d), and $R(\varphi) = R(\varphi')$ by (c). □

Remark 3.4 (Relation to the crystallographic criteria)**.** The image $f(A) \cong A/[P,P]$ is an extension of W by the lattice $2\mathbb{Z}^{\mathcal{T}}$. If $Z(W) = 1$, it is a crystallographic group with holonomy representation ρ. In that case Theorem 3.3 is essentially the Dekimpe–Penninckx criterion [11, Cor. 3.10], [10, Thm. 3.3]: $R = \infty$ iff $\det(1 - AD) = 0$ for some holonomy element A. It is applied to $f(A)$, with the multiplicity-one device of [8, Thm. 4.7] used to produce the eigenvector. If $Z(W) \neq 1$, the holonomy action has kernel $Z(W)$, so $f(A)$ is not crystallographic in the strict sense. Working directly with P avoids this issue.

Remark 3.5.

- If π works for θ, then $g\pi$ works for $\iota_g\theta$, since $\{hg\pi\} = \{h\pi\}$. So one θ per coset of $\mathrm{Inn}(\rho(W))$ suffices.
- For inner θ take $\pi = 1$. Then (i) holds for any good U on which some reflection acts nontrivially. This is automatic for a nontrivial good U when all reflections are conjugate: the kernel of the action is normal, so if it contained one reflection it would contain all of them.
- If $\theta(\rho(\mathcal{T})) = \rho(\mathcal{T})$, then $\pi(r) = \rho^{-1}\theta\rho(r)$ realises θ.

4. The exceptional types

4.1. **Induced automorphisms.** Let W be finite irreducible. Let $R(W) \leq \mathrm{Aut}(W)$ be the reflection-preserving automorphisms and $\mathrm{Gr}(W)$ the graph automorphisms. A *central twist* $w \mapsto w\,c(w)$, with

$c\colon W \to Z(W)$ a homomorphism, induces the identity on $W/Z(W)$. Franzsen and Howlett prove the following ([21, Thm. 31 and the discussion preceding it]; numbering of the authors' preprint, the version we read).

- For W of type $B_n, D_n, E_6, E_7, E_8, F_4, H_3, H_4$, $\mathrm{Aut}(W) = R(W)\mathcal{A}(W)$, where $\mathcal{A}(W)$ consists of central twists.
- $R(W) = \mathrm{Inn}(W)\,\mathrm{Gr}(W)$ except for H_3 and H_4, where $[R(W) : \mathrm{Inn}(W)] = 2$.

In [21] the proof of Theorem 31 is given as an outline, based on the sizes of the conjugacy classes of involutions. (In item (9) of that theorem, for F_4, "α_χ" should read α_ζ.) For the six exceptional types, Appendix B gives independent control of all of $\mathrm{Aut}(\rho(W))$: by exhaustive computation for E_6, E_7, F_4, H_3, H_4, and by hand for E_8.

Lemma 4.1. *Let $\varphi \in \mathrm{Aut}(A)$, with A of type X. Then $\theta_\varphi \in \mathrm{Inn}(\rho(W))\,\Theta_X$, where:*

- $\Theta_X = \{1\}$ *for* $X = E_6, E_7, E_8, B_n$ *($n \geq 3$);*
- $\Theta_{F_4} = \{1, \theta_L\}$, $\Theta_{H_3} = \{1, \theta_\Sigma\}$, $\Theta_{H_4} = \{1, \theta_\Psi\}$;
- $\Theta_{D_n} = \{\theta_\gamma : \gamma \in \mathrm{Gr}(D_n)\}$.

Every element of Θ_X maps $\rho(\mathcal{T})$ onto itself, so it is realised by a permutation of $\mathcal{T}$.

Proof. Central twists act trivially on $\rho(W)$, so θ_φ is induced by an element of $R(W)$.

- For E_6 (and D_n, n odd) the graph automorphism is induced by conjugation by w_0, hence inner. For E_7, E_8 and B_n, Gr is trivial.
- For F_4, conjugation by the similarity $L(x) = (x_1 + x_2, x_1 - x_2, x_3 + x_4, x_3 - x_4)$ satisfies $Ls_\alpha L^{-1} = s_{L\alpha}$. It is reflection-preserving and exchanges the two reflection classes, so it is not inner. Since $[R(W) : \mathrm{Inn}(W)] \leq |\,\mathrm{Gr}(W)| = 2$, it represents the nontrivial coset.
- For H_3, resp. H_4, conjugation by Σ, resp. Ψ (Propositions 4.3 and 4.6) is reflection-preserving and induces a non-inner automorphism of $\rho(W)$. It therefore represents the nontrivial coset of $\mathrm{Inn}(W)$ in $R(W)$.

Finally, $\theta_r(\rho(\mathcal{T})) = \rho(\mathcal{T})$ for $r \in R(W)$, by definition of $R(W)$. □

4.2. **Explicit good constituents.**

Proposition 4.2 (E_6, E_7, E_8)**.** *Let $A_\perp$ be the adjacency matrix of the orthogonality graph on root lines.*

(a) $\rho(W)$ has exactly three orbitals on $\mathcal{T} \times \mathcal{T}$: equality, orthogonality and non-orthogonality.
(b) The graph is strongly regular with parameters $(36, 15, 6, 6)$, $(63, 30, 13, 15)$, $(120, 63, 30, 36)$. Its nontrivial eigenvalues are $3^{15}, (-3)^{20}$; $3^{35}, (-5)^{27}$; $3^{84}, (-9)^{35}$ (multiplicities as exponents).
(c) The two nontrivial eigenspaces are good, and every reflection acts nontrivially on them.

Proof. By Steinberg's theorem [33], the stabiliser of a root, or of a pair of roots, is the Weyl group of the perpendicular root subsystem. The perpendicular subsystems are listed below. The root counts and ranks were checked by machine (Appendix A). Each pair (number of roots, rank) in the table occurs for exactly one simply-laced root system, irreducible or not, of that rank, so the counts determine the types. These are the classical facts that the centraliser of a reflection in $W(E_8)$, $W(E_7)$, $W(E_6)$ involves $W(E_7)$, $W(D_6)$, $W(A_5)$.

	$\alpha^\perp$	$\{\alpha, \gamma\}^\perp$, $\alpha \perp \gamma$	$\{\alpha, \beta\}^\perp$, $(\alpha, \beta) = 1$
E_6	A_5 (30, rank 5)	A_3 (12, rank 3)	A_2^2 (12, rank 4)
E_7	D_6 (60, rank 6)	D_4A_1 (26, rank 5)	A_5 (30, rank 5)
E_8	E_7 (126, rank 7)	D_6 (60, rank 6)	E_6 (72, rank 6)

W is transitive on roots. Inside $W(\alpha^\perp)$, the orbits are as follows.

- The orbit of an orthogonal root has size $720/24 = 30$, $23040/384 = 60$, $2903040/23040 = 126$, i.e. all of $\alpha^\perp$.
- The orbit of a root β with $(\alpha, \beta) = 1$ has size $720/36 = 20$, $23040/720 = 32$, $2903040/51840 = 56$, i.e. all such roots.

Passing to lines gives (a). Then k, λ, μ are half the root counts in the table, and (b) follows from the standard formulas for strongly regular graphs.

By (a) the commutant is spanned by I, $A_\perp$ and J. These act by scalars on the nontrivial eigenspaces, which are rational because the eigenvalues are integers. So Lemma 3.2(i) applies, and Remark 3.5 gives the last assertion. □

Proposition 4.3 (H_3)**.** *(a) $\rho(W) \cong A_5$. The 15 root lines form 5 orthogonal triples (*frames*), which correspond to the Klein four-subgroups.*

(b) The frame module U (functions constant on frames, with sum 0) is a good 4-dimensional constituent. More precisely, writing irreducible A_5-modules by their dimensions,

$$\mathbb{Q}^{\mathcal{T}} \cong 1 \oplus U \oplus 5 \oplus 5.$$

(c) For $\Sigma = \varsigma \circ \sigma$, where σ is Galois conjugation $\sqrt{5} \mapsto -\sqrt{5}$ on coordinates and ς swaps the last two coordinates, $\Sigma\Phi = \Phi$ and $\Sigma s_\alpha \Sigma^{-1} = s_{\Sigma\alpha}$. The induced π_Σ is not inner on $\rho(W)$.

(d) For every outer θ there is h such that $h\pi$ acts on the frames as a transposition. It then has eigenvalues $+1$ (multiplicity 3) and -1 (multiplicity 1) on U.

Proof. (a) Commuting reflections correspond to orthogonal lines, and the 15 involutions of A_5 lie in 5 Klein four-subgroups, permuted as A_5 permutes 5 points.

(b) U is the augmentation module of a 2-transitive action, hence absolutely irreducible. Since $\mathbb{Q}^{\mathcal{T}} = \mathrm{Ind}_{V_4}^{A_5} 1$, the multiplicity of an irreducible χ is $\dim \chi^{V_4}$. This equals $1, 0, 0, 1, 2$ for χ of degree $1, 3, 3', 4, 5$, which gives the displayed decomposition.

(c) Σ satisfies $(\Sigma x, \Sigma y) = \sigma(x, y)$. θ_Σ is not inner: on the rotation subgroup $W^+ \cong \rho(W)$ it Galois-conjugates the character of $\mathbb{R}^3$ on elements of order 5, while inner automorphisms preserve characters. This is also checked by machine.

(d) An outer automorphism of A_5 is conjugation by an odd permutation, and π permutes the frames accordingly. So $\{h\pi\}$ contains a transposition. In fact π_Σ itself is a transposition on frames. □

Proposition 4.4 (F_4)**.** *Each class of 12 root lines (long, short) splits into 3 orthogonal frames of 4 lines. Let $U_{\mathrm{long}}, U_{\mathrm{short}}$ be the corresponding frame modules (sum zero). Then:*

(a) both are good and 2-dimensional;

(b) long reflections act on U_{long} as transpositions of frames and trivially on U_{short}, and vice versa;

(c) π_L exchanges U_{long} and U_{short}, and $\pi_L^2 = 1$;

(d) for a long simple reflection s, $(s\pi_L)^2 = s \cdot \pi_L s \pi_L$ has eigenvalue -1 on U_{short}, and π_L^2 gives eigenvalue $+1$.

Hence case (ii) of Theorem 3.3 holds for θ_L.

Proof. The long roots $\pm e_i \pm e_j$ form a D_4, whose frames correspond to the pairings $ij|kl$. The short lines e_i form one frame. The lines $\frac{1}{2}(\pm 1, \pm 1, \pm 1, \pm 1)$ form two frames, according to the parity of the number of minus signs. Each frame system is permuted through all of S_3, so each frame module is the standard module of S_3.

Multiplicity one (Lemma 3.2(iii)):

- the stabiliser of the long line $e_1 + e_2$ fixes one long frame and contains $s_{e_1 - e_2}$, which swaps the other two, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 1$;

- the stabiliser of the short line e_1 contains $\mathrm{Sym}\{2,3,4\}$, which induces S_3 on pairings, so $\dim U_{\mathrm{long}}^{\mathrm{Stab}} = 0$;
- U_{short} follows by applying π_L.

(b) Check on $s_{e_1-e_2}$ and s_{e_1}, and use conjugacy.
(c) L maps orthogonal frames to orthogonal frames and long to short lines, and $L^2 = 2I$.
(d) $\pi_L s \pi_L$ is a short reflection. □

Lemma 4.5 (Icosian model of H_4)**.** *Halve the* 120 *roots of* H_4 *(Appendix A) and read the first coordinate as the real part. These form a group* I *of unit quaternions (the binary icosahedral group), and* $K = I/\{\pm 1\} \cong A_5$*. Identify* $\mathcal{T}$ *with* K*. Then:*

(a) the reflection in a unit root r *is* $x \mapsto -r\bar{x}r$*, and it acts on* K *by* $x \mapsto rx^{-1}r$*;*
(b) $\rho(W) = \{x \mapsto ax^{\pm 1}b : a, b \in K\} \cong (K \times K) \rtimes C_2$*.*

Proof. (a) For $|r| = 1$, $r\bar{x}r = 2(x,r)r - x$, using $\bar{x}r + \bar{r}x = 2(x,r)$.
(b) By (a), $\rho(W)$ is contained in the right-hand group, which has order $7200 = |W|/2 = |\rho(W)|$. The closure of I and the formulas are also checked by machine. □

Proposition 4.6 (H_4)**.** *Let* V *be the* 4*-dimensional augmentation module of* $K \cong A_5$ *acting on its five Klein four-subgroups, and let* $U \subseteq \mathbb{Q}^K$ *be the span of the matrix coefficients* $x \mapsto \mathrm{tr}(X\lambda_V(x))$*, so* $U \cong \mathrm{End}(V)$ *has dimension* 16*.*

(a) U *is good, and* $\mathbb{Q}^{\mathcal{T}} \cong 1 + (3 \otimes 3) + (3' \otimes 3') + 16 + 25$*.*
(b) Let $\Psi = \Omega \circ \sigma$*, where* σ *is Galois conjugation and* Ω *is the rotation* $j \mapsto k$*,* $k \mapsto -j$*. Then* Ψ *is a ring automorphism of* $\mathbb{H}_{\mathbb{Q}(\sqrt{5})}$ *with* $\Psi(I) = I$*. The induced permutation* π_Ψ *normalises* $\rho(W)$*, preserves reflections, and is not inner.*
(c) For every outer θ *there is* h *such that* $h\pi$ *acts on* U *as conjugation by a transposition. This has eigenvalues* $+1$ *(multiplicity* 10*) and* -1 *(multiplicity* 6*).*

Proof. (a) By Wedderburn, $\mathbb{C}[K] = \bigoplus_{V'} V' \otimes V'^*$ as a $K \times K$-module, each summand irreducible and occurring once. The summand for V is U, which is rational and stable under inversion. The decomposition follows from the irreducible degrees $1, 3, 3, 4, 5$ of A_5.

(b) Ψ commutes with quaternion conjugation, so $\Psi s_r \Psi^{-1} = s_{\Psi r}$. Closure $\Psi(I) = I$ is checked exactly. If θ_Ψ were inner, then $\pi_\Psi \in \rho(W)$, because the centraliser of $\rho(W)$ in $\mathrm{Sym}(K)$ is trivial. Since π_Ψ fixes 1, it would be $x \mapsto ax^{\pm 1}a^{-1}$. The sign $-$ is impossible, since π_Ψ is an automorphism of the non-abelian group K. The sign $+$ is impossible, since conjugation preserves real parts up to sign, whereas Ψ maps an element of order 5 with real part $\tau/2$ to one with real part $-\tau^{-1}/2$ (here $\tau = \frac{1+\sqrt{5}}{2}$).

(c) We argue in three steps.

(i) By Lemma 4.1 and Remark 3.5, for an outer θ we may take $\pi \in \rho(W)\pi_\Psi$. So $\{h\pi : h \in \rho(W)\} = \{h\pi_\Psi\}$.
(ii) By (b), $\pi_\Psi = \psi$ is an outer automorphism of K.
(iii) Since $\mathrm{Aut}(A_5) \cong S_5$, acting on the five Klein four-subgroups, ψ is conjugation by an odd permutation δ'. For $a \in K = A_5$, the element $h = (x \mapsto axa^{-1})$ lies in $\rho(W)$, and $h\psi$ is conjugation by $a\delta'$. Choose a with $a\delta' = \delta$ a transposition.

The action of K on V extends to S_5, with $\lambda_V(\delta)$ having eigenvalues $1, 1, 1, -1$. Since $h\pi_\Psi$ acts on $U \cong \mathrm{End}(V)$ as $X \mapsto \lambda_V(\delta) X \lambda_V(\delta)^{-1}$, the eigenvalue $+1$ has multiplicity $9 + 1$ and -1 has multiplicity 6. □

Proof of the Main Theorem. By Lemma 4.1 and Remark 3.5, it suffices to treat Θ_X.

- E_n: only the inner class occurs. Proposition 4.2 supplies U, and Remark 3.5 applies, since all reflections are conjugate.

- H_3, H_4: the inner class is covered by Remark 3.5 (all reflections are conjugate and U is nontrivial). The outer class is covered by Propositions 4.3 and 4.6: there $\pi U = U$, since U is the only good constituent of its dimension, and case (i) applies.
- F_4: there are two classes of reflections, so the inner class is treated explicitly. Take $\pi = 1$ and $U = U_{\mathrm{long}}$; then $h = 1$ gives $+1$, and a long reflection gives -1 by Proposition 4.4(b), so case (i) holds. The outer class θ_L is covered by Proposition 4.4, case (ii).
- D_5: Proposition 5.2.

Apply Theorem 3.3. □

type	$\lvert\mathcal{T}\rvert$	$\lvert\rho(W)\rvert$	Θ_X	good constituent used (dim)	outer witness
E_6	36	51840	1	**SRG** eigenspace (15; also 20)	—
E_7	63	1451520	1	**SRG** eigenspace (35; also 27)	—
E_8	120	348364800	1	**SRG** eigenspace (84; also 35)	—
F_4	24	576	$\{1, \theta_L\}$	$\mathbf{U_{long}}, \mathbf{U_{short}}$ (2+2)	swap: π_L^2, $(s\pi_L)^2$
H_3	15	60	$\{1, \theta_\Sigma\}$	**frame** module (4)	eigen-multiplicities 3, 1
H_4	60	7200	$\{1, \theta_\Psi\}$	End(**V**) (16)	eigen-multiplicities 10, 6

TABLE 1. The data used for the Main Theorem. The dimensions agree with independent computer certificates (Appendix A).

5. Uniform proofs for the classical types

Except for D_5, these results are known [5, 30]. We include them to illustrate the method, and because for D_5 the argument gives what appears to be the first proof (Remark 5.3).

Proposition 5.1 (A_n, $n \geq 2$)**.** *$A(A_n)$ has property R_∞.*

Proof. Here $\rho(W) = S_{n+1}$ acts on the transpositions, i.e. on the 2-subsets. By Young's rule $\mathbb{Q}^{\mathcal{T}} = S^{(n+1)} \oplus S^{(n,1)} \oplus S^{(n-1,2)}$ (for $n = 2$ the last summand is absent). So $S^{(n,1)}$ is good. For $n+1 \neq 6$, $\mathrm{Aut}(S_{n+1}) = \mathrm{Inn}$. For $n+1 = 6$, an outer automorphism maps $(12) \mapsto (12)(34)(56)$, but $\chi_{\mathcal{T}}$ takes the values 7 and 3 there, so it is not of the form θ_φ (Corollary 2.5). Apply Remark 3.5. □

For B_n and D_n let $u_{ij} = e_{[e_i+e_j]} - e_{[e_i-e_j]}$ $(i < j)$ and $U^- = \mathrm{span}\{u_{ij}\}$. Let $E \leq W$ be the group of sign changes: all of them for B_n, the even ones for D_n.

Proposition 5.2 (B_n, $n \geq 3$; D_n, $n \geq 5$)**.** *U^- is good, and $A(B_n)$ and $A(D_n)$ have property R_∞.*

Proof. Weights. E acts on u_{ij} by $\chi_{ij}(\varepsilon) = \varepsilon_i\varepsilon_j$. These characters are nontrivial and pairwise distinct on E. For D_n this needs $n \geq 5$: on even sign changes $\chi_{ij} = \chi_{kl}$ iff $\varepsilon_i\varepsilon_j\varepsilon_k\varepsilon_l$ is trivial or the product of all signs.

Goodness. E fixes $e_{[e_i+e_j]} + e_{[e_i-e_j]}$ and the short lines. So each χ_{ij} occurs once in $\mathbb{Q}^{\mathcal{T}}$. Since $S_n \leq W$ permutes the weights transitively, $U^- \otimes \mathbb{C}$ is irreducible.

Automorphisms. By Lemma 4.1, θ_φ is inner for B_n, and inner up to the diagram involution for D_n (for odd n the latter is inner too). For the inner class, $\pi = 1$ works: the reflection s_{e_n} (B_n), resp. $s_{e_{n-1}-e_n}$ (D_n), acts nontrivially on U^-. The diagram involution is realised by negating e_n, which acts on u_{ij} by -1 if $n \in \{i, j\}$ and by $+1$ otherwise. So case (i) holds with $h = 1$. □

Remark 5.3 (Type D_5)**.** Proposition 5.2 with $n = 5$ proves R_∞ for $A(D_5)$. Its inputs are [7], [21] (here $\mathrm{Aut}(W(D_5)) = R(W) \cong W$ by [21, Thm. 31(3)]) and the weight argument.

There is also a short derivation from Paris–Soroko's description $\mathrm{Aut}(A[D_5]) = \mathrm{Inn}(A) \rtimes \langle\chi\rangle$, with χ the inversion of the generators [28, Thm. 1]:

- $R(\iota_g) = R(\mathrm{id}) = \infty$ by Lemma 2.1(c), since $A^{ab} \cong \mathbb{Z}$ and Lemma 2.1(a),(b) give infinitely many conjugacy classes;
- $R(\iota_g \circ \chi) = R(\chi) = \infty$ by Remark 5.6(b).

This deduction is nearly immediate once [28, Thm. 1] is available, but we have not found it written down. [30] states D_5 as a conjecture, and [28] does not mention R_∞. We therefore record D_5 as part of the Main Theorem.

Proposition 5.4 (D_4)**.** *In type D_4:*

- *U^- is reducible: $J(u_{ij}) = u_{kl}$ ($\{k,l\}$ complementary) commutes with $W(D_4)$, and U^- is the sum of two good 3-dimensional constituents;*
- *the frame module U on the 3 orthogonal frames $\{12|34, 13|24, 14|23\}$ is good of dimension 2;*
- *$A(D_4)$ has property R_∞.*

Proof. The E-fixed part of $\mathbb{Q}^{\mathcal{T}}$ is the permutation module on the 6 pairs, namely $1+3+2$ for S_4. Its 2-dimensional summand is U, which does not occur in U^-. Every θ_φ preserves reflections, so π permutes the frames. W induces all of S_3 on frames, so some $h\pi$ is a transposition there, with eigenvalues ± 1 on U. □

Remark 5.5 (Dihedral types)**.** Let m be odd. Then $Z(W) = 1$, so ρ is faithful and injective on $\mathcal{T}$, and $\mathbb{Q}^{\mathcal{T}} \cong \bigoplus_{d|m} U_d$ with $\dim U_d = |(\mathbb{Z}/d\mathbb{Z})^\times|$, and U_d is absolutely irreducible iff $d \in \{1, 3\}$.

- If $3 \nmid m$ there is no nontrivial good constituent, and Theorem 3.3 cannot be applied. For m prime, Schur's lemma gives only a unit of $\mathbb{Q}(\cos \frac{2\pi}{m})$.
- If $3 \mid m$, then U_3 is the unique good constituent of dimension 2, so $\pi U_3 = U_3$. The group $\rho(W)$ acts on U_3 as $S_3 \leq O(2)$. Since π is orthogonal and normalises S_3, it acts as an element of $S_3 \times \{\pm 1\}$. Hence some $h\pi$ acts as $\pm$(a reflection), with eigenvalues ± 1. So Theorem 3.3 applies, using [7] and that every automorphism of W preserves reflections.

This was confirmed by computation for odd $m \leq 21$. We have not analysed even m. In any case R_∞ for $I_2(m)$ is known [5].

Remark 5.6 (Two elementary observations)**.** (a) If φ fixes the generator $z \in \{\Delta, \Delta^2\}$ of $Z(A)$ [4, 12], then φ_* fixes the nonzero image of z in $H_1(A)$, so $R(\varphi) = \infty$. Only centre-inverting automorphisms need the criterion.

(b) For any Artin group that is not free abelian, the inversion $s \mapsto s^{-1}$ has $R = \infty$. Choose $m_{st} \geq 3$. Via f, $\max_r |v_r - v_{srs}|$ is an invariant of twisted classes with W-part s, and it separates the elements $t^{2k}s$. Cf. [29, Thm. 1], [24].

Appendix A. Computational cross-checks

The following were checked by exact arithmetic (over $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{5})$). Programs, commands and expected outputs are listed in `code/README.md` of the accompanying repository.

Coordinates. For E_8 we use Bourbaki's simple roots in $\mathbb{R}^8$ with basis $e_1, \dots, e_8$:

$$\alpha_1 = \tfrac{1}{2}(e_1 + e_8) - \tfrac{1}{2}(e_2 + \cdots + e_7), \qquad \alpha_2 = e_1 + e_2, \qquad \alpha_k = e_{k-1} - e_{k-2} \quad (3 \leq k \leq 8).$$

E_7 and E_6 use $\alpha_1, \dots, \alpha_7$, resp. $\alpha_1, \dots, \alpha_6$, in $\mathbb{R}^8$. The other types:

- F_4: $e_2 - e_3$, $e_3 - e_4$, e_4, $\frac{1}{2}(e_1 - e_2 - e_3 - e_4)$;
- A_n, B_n, D_n: standard simple roots;
- H_3: roots $\pm 2e_i$ and the even permutations of $(\pm\tau, \pm 1, \pm\tau^{-1})$, with $\tau = \frac{1+\sqrt{5}}{2}$;
- H_4: roots $\pm 2e_i$, $(\pm 1, \pm 1, \pm 1, \pm 1)$ and the even permutations of $(\pm\tau, \pm 1, \pm\tau^{-1}, 0)$.

Checks.

(1) Proposition 4.2: SRG parameters, three orbitals, eigenspace dimensions, perpendicular root counts and ranks.

(2) Propositions 4.3–4.6 and Lemma 4.5:
- frames, and character norms $\langle\chi,\chi\rangle = \langle\chi,\chi_{\mathcal{T}}\rangle = 1$;
- the images of W on frames (A_5, S_3);
- the action of π_Σ, π_L, π_Ψ, with eigenspace dimensions $(3,1)$, $(1,1)$, $(10,6)$;
- the closure of I, the reflection formula, and $\rho(W) = \{x \mapsto ax^{\pm1}b\}$.

(3) Propositions 5.2 and 5.4:
- the weight argument for B_3–B_6 and D_4–D_7, including its failure for D_4;
- the operator J and the frame module of D_4;
- witnesses for all six graph automorphisms of D_4.

(4) Earlier, independent certificates for the six exceptional types:
- commutant certificates;
- character certificates written as fresh code;
- witnesses for abstractly computed outer automorphisms.

They give the same dimensions $15, 20$; $27, 35$; $35, 84$; 4; $2+2$; $16, 25$.

(5) The criterion (for automorphisms preserving $\chi_{\mathcal{T}}$) checked uniformly for A_2–A_7, B_3–B_7, D_4–D_7, odd $I_2(m)$ with $m \le 21$, and the exceptional types other than E_8.

Appendix B. Corroboration: all automorphisms of $W/Z(W)$

The argument above uses only automorphisms of $\rho(W)$ induced from $\mathrm{Aut}(W)$. The Lean formalisation assumes the stronger statement `hOut` about all of $\mathrm{Aut}(\rho(W))$:

- $\mathrm{Out}(\rho(W)) = 1$ for E_6, E_7, E_8;
- $|\mathrm{Out}(\rho(W))| = 2$ for H_3, H_4, F_4.

For E_6, E_7, H_3, H_4, F_4 this was verified by exhaustive computation in two ways: brute force over generator images, and involution classes followed by orbits of Coxeter tuples. For E_8 the following argument by hand gives it.

Lemma B.1. *Let Φ be an irreducible simply-laced root system of rank ≥ 3 (so its diagram is a tree). Then every $\theta \in \mathrm{Aut}(\rho(W))$ with $\theta\rho(\mathcal{T}) = \rho(\mathcal{T})$ is inner up to a graph automorphism.*

Proof. Write $\theta\rho(s_i) = \rho(s_{\beta_i})$; this uses that ρ is injective on $\mathcal{T}$. Orders of products agree in W and in $\rho(W)$. Indeed, for roots $\beta \ne \pm\gamma$ the element $s_\beta s_\gamma$ is a plane rotation fixing a complement of dimension ≥ 1, so no nontrivial power of it lies in $Z(W) \subseteq \{\pm1\}$; in particular $s_i s_j \ne -1$. Hence $(\beta_i, \beta_j) = 0$ or ±1 according to the diagram. Choose the signs along the tree. Then $g\colon \alpha_i \mapsto \beta_i$ is an isometry, and $g\Phi = (gWg^{-1})\{\beta_i\} \subseteq \Phi$. So $g\{\alpha_i\}$ is a simple system. As W is simply transitive on simple systems, $g = w\gamma$ with $w \in W$ and γ an isometry permuting $\{\alpha_i\}$, i.e. $g \in W \rtimes \mathrm{Gr} = \mathrm{Aut}(\Phi)$ [32]. Compare [21, Thm. 18, Cor. 19]. □

Proposition B.2. *Every automorphism of $W(E_8)/\{\pm1\}$ is inner.*

Proof. $G^+ = W^+/\{\pm1\}$ is the commutator subgroup, of index 2. An involution outside G^+ lifts to w with $w^2 = 1$ and $\det w = -1$; if $w^2 = -1$ then $\det w = 1$. By [6, Lemma 5] (provable via Steinberg's theorem), w is a product of $k = \dim\ker(w+1)$ reflections in mutually orthogonal roots, and k is odd. Replacing w by $-w$, we may take $k \in \{1, 3\}$. For $k = 3$, the (-1)-eigenspace of w is spanned by the three roots, and contains only the 6 roots $\pm\beta_i$: a root $\sum c_i\beta_i$ has $2c_i = (\cdot, \beta_i) \in \{0, \pm1, \pm2\}$ and $\sum c_i^2 = 1$. So w determines the triple, and $-w$ (with $k = 5$) is not of this form.

The products for $k = 3$ form one class of size $240 \cdot 126 \cdot 60/48 = 37800$, by transitivity along $E_8 \supset E_7 \supset D_6$. So the reflections form the unique class of size 120 outside G^+. Hence every automorphism preserves $\rho(\mathcal{T})$, and Lemma B.1 applies. □

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